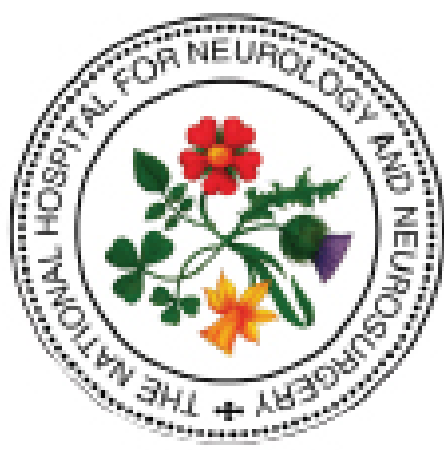




METARAMINOL

INDICATIONS FOR USE

Metaraminol is the vasoconstrictor of choice for the short-term management of acute hypotension and can be administered by peripheral intravenous catheter. Onset of action: 1–2 minutes, Duration of action: 20–60 minutes, Half-life: minutes.

Metaraminol is an intravenous vasopressor that increases both systolic and diastolic BP. The pressor effect of a single dose of metaraminol lasts from about 20 minutes up to one hour. Its onset is around one or two minutes after direct intravenous injection. The vasopressor effects taper off when therapy is stopped.

It can be given IV peripherally to treat hypotension or support BP and can be given as a **BOLUS**.

METARAMINOL DOES NOT NEED TO BE 'DOUBLE PUMPED'

- Short period of disconnection will not cause a drop in BP
- An existing syringe can be disconnected and a new one replaced within the space of 3-4min without a significant drop in BP

FACTORS THAT WILL DETERMINE SWITCHING TO NORADRENALINE

- If the need for vasopressors is going to exceed 48 hours
- If the metaraminol dose requirement is high and increasing, needing frequent syringe changes and high volumes
- If a central line is inserted for other reasons, metaraminol should be switched to noradrenaline

COMPATABILITY and STABILITY

Compatible with 0.9% Sodium chloride / 5% Dextrose / Hartmann's Solution / 4% dextrose / 0.18% saline

- Metaraminol should be given through a dedicated peripheral line.
- The standard dilution is to produce a solution containing 0.5mg/ml (2 ampoules of 10mg/mL diluted in 0.9% sodium chloride to make a total of 40ml)
- The reconstituted solution should be used immediately but is stable in the syringe for up to 24 hours - *discard unused solutions after 24 hours*

CAUTIONS and CONTRAINDICATIONS

- Use with caution in patients taking digoxin as the combination may provoke arrhythmia
- Monoamine oxidase inhibitors and tricyclic antidepressants may potentiate the action of metaraminol ∴ infusion should be titrated to effect with care

SIDE EFFECTS

- Sinus or ventricular tachycardia
- Bolus dose can cause marked hypertension and bradycardia
- Abscess formation and tissue necrosis at site of injection
- Malaria relapse in those patients with previous exposure
- Conscious patients may complain of headache

METARAMINOL for SHORT-TERM MANAGEMENT of ACUTE HYPOTENSION

PRESENTATION

1 Ampoule of metaraminol tartrate contains **10mg in 1ml** solution for injection

STEP 1

ADMINISTRATION

Dilute **20mg** of metaraminol (2 ampoules of 10mg/mL) with 0.9% sodium chloride to produce a total solution of 40ml (20mg/40mL = **0.5mg/mL**)

Deliver by syringe pump peripherally (via **dedicated peripheral intravenous line**)

STEP 2

BOLUS DOSE

If required, an initial **bolus dose of 1ml (0.5mg)** may be given by medical staff

STEP 3

INFUSION

Continuous infusion of 20mg/40mL = **0.5mg/mL** titrated to BP target

Start the infusion at rate of **2ml/h = 1mg/h**

Increase infusion rate by 2ml/h every 15 minutes until BP target reached - *If requiring >20ml/h inform medical staff*

STEP 4

WEANING

When target BP is consistently achieved with only **1-2ml/h** of metaraminol, the infusion can be safely discontinued

The primed line should be left connected on the syringe driver in case the infusion needs to be restarted within the following hour

If BP target still maintained after **1 HOUR** the metaraminol infusion can be disconnected and discarded

The cannula should then be flushed to remove traces of residual drug

UKCPA Clinical Pharmacy Association
Intensive Care Society
Standard Medication Concentrations for Continuous Infusions in Adult Critical Care

The Intensive Care Society supports the adoption of standard concentrations and endorses the recommendations of a multi-professional group who have published a list of concentrations with wide acceptance by critical care for this purpose [1,2,3].

Standardising concentrations represents a significant step towards improving both patient safety and efficient use of resources within critical care. It also facilitates the production of a national medicines guide to provide the end user with the information necessary to safely administer such medications [4].

The adoption of these is recommended, not mandated. It is anticipated that pharmaceutical manufacturers will begin to prepare ready to use and ready to administer products based on this list.

Medication	Concentration	Example Infusion Concentration	Central or Peripheral
Morphine	1mg/mL	50mg in 50mL	C / P
	2mg/mL	100mg in 50mL	C / P
Fentanyl	50micrograms/mL	2.5mg in 50mL	C / P
Alfentanil	500micrograms/mL	25mg in 50mL	C / P
Remifentanyl	50micrograms/mL	2mg in 40mL	C / P
	100micrograms/mL	5mg in 50mL	C / P
Midazolam	1mg/mL	50mg in 50mL	C / P
	2mg/mL	100mg in 50mL	C / P
Clonidine	15micrograms/mL	750micrograms in 50mL	C / P
Dexmedetomidine	4micrograms/mL	200micrograms in 50mL	C / P
	8micrograms/mL	400micrograms in 50mL	C / P
	80micrograms/mL	4mg in 50mL	C / P
Adrenaline	160micrograms/mL	8mg in 50mL	C / P
	320micrograms/mL	16mg in 50mL	C
	16micrograms/mL	4mg in 250mL	C / P
Noradrenaline	80micrograms/mL	4mg in 50mL	C
	160micrograms/mL	8mg in 50mL	C
	320micrograms/mL	16mg in 50mL	C
Vasopressin (tergiversol)	0.4units/mL	20units in 50mL	C / P
Dobutamine	5mg/mL	250mg in 50mL	C
		500mg in 100mL	C

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